Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau

TEAM ID	SWTID2026-3387
PROJECT TITLE	Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau
MAX MARKS	1 MARK

CHAPTER – V

PROJECT PLANNING AND SCHEDULING

5.1 – Project Milestones & Tasks

1. Data Collection & Extraction from Database

The first phase of the project focuses on collecting reliable housing market data from various sources such as CSV files, Microsoft Excel spreadsheets, or SQL databases. The dataset contains important information including sale prices, house area, number of bedrooms and bathrooms, year built, basement area, property condition, and location details. After collecting the data, Tableau is connected to the selected data source to extract the required records. During this process, the dataset is verified for completeness and accuracy to ensure that only valid and consistent information is used for analysis. Successful data extraction provides the foundation for meaningful visualizations and accurate insights into housing market trends.

2. Data Preparation

Data preparation is one of the most important stages of the project because raw datasets often contain missing values, duplicate records, inconsistent formats, and incorrect data types. In this phase, the housing dataset is cleaned by removing duplicate entries, handling missing values, correcting data formats, and standardizing column names. Additional calculated fields such as house age and price per square foot may also be created to support advanced analysis. The prepared dataset becomes organized, accurate, and ready for visualization, ensuring that the final dashboard produces reliable and meaningful result

3. Data Visualization

After the dataset is cleaned, Tableau is used to create various visualizations that help identify housing market patterns and relationships. Different chart types are selected based on the type of analysis required. Bar charts compare sale prices across categories, line charts display market trends over time, scatter plots examine relationships between house features and prices, histograms show data distribution, and heat maps highlight density patterns. These visualizations simplify complex housing data, allowing users to quickly understand property values, market behavior, and feature comparisons through interactive graphical representations.

4. Dashboard

The dashboard combines multiple visualizations into a single interactive interface that allows users to explore housing market information efficiently. It includes charts, filters, legends, KPIs, and tooltips that enable users to analyze property prices based on different characteristics such as location, number of bedrooms, house size, and construction year. Interactive filters allow users to customize the displayed information without modifying the original dataset. The dashboard is designed with a user-friendly layout that supports quick decision-making and provides a comprehensive overview of housing market performance.

5.Storyboard

The Tableau Storyboard presents the complete analytical journey of the project by arranging multiple dashboards and worksheets into a logical sequence. It begins with an introduction to the dataset, followed by visual analysis of housing prices, property characteristics, market trends, and final observations. Each story point explains a specific aspect of the analysis and guides users toward understanding the overall findings. The storyboard transforms individual visualizations into a meaningful presentation that effectively communicates insights to both technical and non-technical audiences.

6. Performance Testing

Performance testing is conducted to ensure that the Tableau dashboard operates efficiently under different conditions. The testing process measures dashboard loading speed, filter response time, calculation accuracy, and overall system performance. Optimization techniques such as reducing unnecessary calculations, limiting excessive filters, and improving data source efficiency are applied to enhance dashboard responsiveness. Performance testing guarantees that users experience smooth navigation and fast interaction while exploring housing market data

7. Web Integration

Once the dashboard is completed, it is integrated into a web application to make the analysis accessible through a browser. The Tableau dashboard is published and embedded into a Flask-based website, allowing users to interact with the visualizations online without requiring Tableau Desktop. The web application is designed to be responsive so that it functions properly across desktops, laptops, tablets, and mobile devices. Browser compatibility and accessibility testing are also performed to ensure a consistent user experience.

8. Project Demonstration & Documentation

The final phase of the project involves preparing complete documentation and demonstrating the developed solution. Documentation includes the project overview, objectives, methodology, dataset description, dashboard design, testing procedures, results, conclusions, and future enhancements. During the project demonstration, the workflow from data collection to dashboard development is explained, followed by a live presentation of the interactive Tableau dashboard and storyboard. The demonstration highlights the project's ability to visualize housing market trends, analyze property features, and support informed decision-making through clear and interactive data visualizations.